

Ali Muqtadir

Email address: sam24@stir.ac.uk | Website: sa-muqtadir.github.io | Address: Department of Psychology, University of Stirling, FK94LA, Stirling, United Kingdom (Work)

Education & Training

PhD Neuroscience | University of Stirling | 01/10/2022 - Current | Stirling, United Kingdom

Level in EQF: 8 | Thesis: The Heading Brain: Characterising the Acute Neurobiological Response to Football Heading

MSc Neuroscience | King's College London | 25/09/2017 - 30/09/2018 | London, United Kingdom

Level in EQF: 7 | Thesis: Effect of transcranial direct current stimulation on probabilistic learning in patients with schizophrenia

BSc (Hons) Psychology | University of Central Punjab | 01/09/2012 - 22/12/2016 | Lahore, Pakistan

Level in EQF: 6

Work experience

Doctoral Researcher | University of Stirling | 01/10/2022 - Current | Stirling, United Kingdom

- My PhD project focused on isolating the acute effects of football heading on cortical excitation-inhibition, neuronal structural integrity, and neural metabolites
- Used 7T MR spectroscopy to investigate heading effects on brain metabolites at the Imaging Centre for Excellence (ICE), Glasgow and Forschungszentrum Jülich
- Explored how heading changes the trajectory blood-based biomarkers of brain injury over 24 hours in a serial sampling study using ultra-sensitive immunoassays at UK DRI labs, University of Edinburgh
- Used concurrent TMS-EEG to investigate excitation-inhibition balance changes following heading exposure
- Collected and contributed heading kinematic data to computational simulations of brain biomechanics following heading impacts in collaboration with the Department of Engineering, University of Edinburgh
- Led and contributed to successful funding applications, ensured compliance with local and external ethics and governance procedures including international cooperation agreements
- Contributed to increasing technical capabilities at Stirling with the first implementation of TMS-EEG and adoption of a low-cost navigated TMS system, and at ICE, Glasgow with the first implementation of an edited MRS sequence on the 7T scanner

Research Associate | Lahore University of Management Sciences | 01/08/2019 - 31/08/2022 | Lahore, Pakistan

- Worked in the areas of consumer behaviour and communication to investigate how 'viral' content spreads online and the factors that mediate the spread of misinformation or 'fake news'
- Contributed to three research projects on online misinformation, social media communication and digital content virality
- Co-authored empirical case studies applying a logistic diffusion model to real-world YouTube virality data for Bhimani et al. (2024)
- Contributed to study design and manuscript drafting for Ali et al. (2022)

Graduate Researcher | Department of Psychosis Studies, King's College London | 01/02/2018 - 31/08/2018 | London, United Kingdom

- Worked on a project investigating the effects of non-invasive brain stimulation on cognitive processes in patients with schizophrenia
- Investigated the effects of transcranial direct current stimulation (tDCS) on probabilistic learning in schizophrenia, contributing to publications in Psychiatry Research and Scientific Reports
- Administered tDCS protocols, structured clinical interviews (PANSS), and cognitive assessments within a clinical research setting

Skills

Human Data Collection | Magnetic Resonance Spectroscopy | Magnetic Resonance Imaging | Electroencephalography | Transcranial Magnetic Stimulation | Transcranial Direct Current Stimulation | Electromyography | Venepuncture | Peripheral IV Cannulation | Cognitive Testing | Neuropsychological Assessments | Structured Clinical Interviews

Programming | R | MATLAB

Publications

[Acute temporal dynamics of brain injury plasma biomarkers following controlled football heading](#)

Muqtadir, S. A., Aston, T., Reid, K. R., Lember, L.-M., Di Virgilio, T. G., Keith, G., Hunt, D., Teixeira-Dias, F., Binkofski, F., Moran, C. N., & Ietswaart, M. (2026). bioRxiv 2026.09.09.749919.

Preprint

[Social media virality: Reaching the tipping point](#)

Bhimani, A., Zainulabdin, K., Ali, K., Ali Muqtadir, S., & Hausken, K. (2024). Journal of International Business Research and Marketing 8(3), 27–41.

[The effects of emotions, individual attitudes towards vaccination, and social endorsements on perceived fake news credibility and sharing motivations](#)

Ali, K., Li, C., Zain-ul-abdin, K., & Muqtadir, S. A. (2022). Computers in Human Behavior 134, 107307.

[Stimulating learning: A functional MRI and behavioral investigation of the effects of transcranial direct current stimulation on stochastic learning in schizophrenia](#)

Orlov, N. D.*, Muqtadir, S. A.*, Oroojeni, H., Averbek, B., Rothwell, J., & Shergill, S. S. (2022). Psychiatry Research 317, 114908.

[The effect of training intensity on implicit learning rates in schizophrenia](#)

Orlov, N. D.*, Sanderson, J.*, Muqtadir, S. A.*, Kalpakidou, A. K., Michalopoulou, P. G., Lu, J., & Shergill, S. S. (2021). Scientific reports 11(1), 6511.

[The effect of football heading on blood biomarkers of brain injury](#)

Muqtadir, S. A., Aston, T., Moran, C., Ietswaart, M. (2025). ISRCTN

Pre-registration

[Impact of Football Heading on TMS-EEG Indicators of Neural Excitation and Inhibition](#)

Muqtadir, S. (2025). OSF

Pre-registration

Certifications

MRI Safety Training for MHRA Category B Staff | NHS Scotland TURAS Learn | 12/07/2026

Peripheral I.V Cannulation | Geopace Training | 12/03/2025

Phlebotomy | Geopace Training | 21/06/2023

Teaching and Supervision

University of Stirling | 01/10/2022 - Current

- Co-supervised a master's project investigating EEG activity patterns and natural environment exposure
- Ran weekly sessions on EEG ERP analysis in Brain Vision Analyzer for work placement students
- Supervised undergraduate research projects and ran tutorials for research design in psychology
- Conducted lab demonstrations for master's students, including practical TMS and EEG sessions
- Facilitated interdisciplinary Crucible workshops as Faculty Officer, bringing together researchers across disciplines
- Carried out marking for undergraduate assignments across multiple modules

Lahore University of Management Sciences | 01/08/2019 - 31/08/2022

- Trained undergraduate researchers in eye-tracking data collection on a Gazepoint system
- Delivered teaching on research methods for undergraduate students

Funding and Awards

2025 - RS MacDonald Facility Access Funding

2022 - SINAPSE Early Career Exchange Fund

2022-2026 - Medical Research Scotland PhD Studentship

2013-2016 - Academic Merit Scholarship, University of Central Punjab